

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12408828
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Tani; Takaharu et al.

Imaging device and endoscope system

Abstract

An imaging device includes a camera unit and a control unit. The camera unit includes an image sensor. The control unit includes a voltage measurement circuit and a voltage adjustment circuit. A first power source voltage is transferred from the control unit to the camera unit by a power source line and is input to the camera unit as a second power source voltage. A video signal or the second power source voltage is output to a video signal line. The voltage measurement circuit is configured to measure a voltage value of the second power source voltage. The voltage adjustment circuit is configured to adjust a value of the first power source voltage based on the value of the second power source voltage.

Inventors: Tani; Takaharu (Hachioji, JP), Ogawa; Keisuke (Hachioji, JP), Harada; Yasunari (Hachioji, JP)

Applicant: OLYMPUS MEDICAL SYSTEMS CORP. (Hachioji, JP)

Family ID: 80846308

Assignee: OLYMPUS MEDICAL SYSTEMS CORP. (Tokyo, JP)

Appl. No.: 18/189627

Filed: March 24, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230225600 A1	Jul. 20, 2023

Related U.S. Application Data

continuation parent-doc WO PCT/JP2020/036527 20200928 PENDING child-doc US 18189627

Publication Classification

Int. Cl.: A61B1/00 (20060101); A61B1/045 (20060101); A61B1/05 (20060101)

U.S. Cl.:

CPC A61B1/05 (20130101); A61B1/045 (20130101); A61B1/00009 (20130101)

Field of Classification Search

CPC: A61B (1/05); A61B (1/045); A61B (1/00009); A61B (1/00004); A61B (1/00025); A61B (1/00027); H04N (5/63); G02B (23/2492)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10778933	12/2019	Shigehisa	N/A	G02B 23/2492

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2008-161427	12/2007	JP	N/A
2011-206333	12/2010	JP	N/A
2016-214381	12/2015	JP	N/A
2016-214571	12/2015	JP	N/A

OTHER PUBLICATIONS

International Search Report dated Nov. 2, 2020, issued in counterpart Application No. PCT/JP2020/036527 with English translation. (4 pages). cited by applicant

Primary Examiner: Neal; Timothy J

Attorney, Agent or Firm: WHDA, LLP

Background/Summary

(1) The present application is a continuation application based on International Patent Application No. PCT/JP2020/036527 filed on Sep. 28, 2020, the content of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to an imaging device and an endoscope system.

Description of Related Art

(2) An endoscope system includes an endoscope (camera unit) and a main body, and the endoscope and the main body are connected to each other by a cable. An imager is mounted in the distal end of the endoscope. A power source voltage used for driving the imager is transferred from the main body to the distal end of the endoscope via the cable. Hereinafter, the power source voltage that has

reached the distal end of the endoscope will be called a distal end voltage.

(3) The power source voltage needs to be adjusted such that the distal end voltage has an appropriate value in order to drive the imager stably. However, the value of the distal end voltage may deviate from a voltage range recommended for an operation of the imager due to factors such as the length of the cable, deviations and fluctuations of the characteristics of the cable, and fluctuations of a current used for driving the imager. Hereinafter, the voltage range recommended for the operation of the imager will be called a recommended voltage range. As a result, problems may occur in driving the imager stably. Particularly, when the cable is long or thin, the influence thereof becomes noticeable and it is highly probable that the value of the distal end voltage deviates from the recommended voltage range.

(4) In the prior art, a high-power source voltage is output to the cable in order for the value of the distal end voltage to fall in the recommended voltage range. However, since the accuracy and the follow-up performance are low, a problem occurs in that the image quality deteriorates and heat is generated. In addition, a problem also occurs in that the thickness of the cable increases in order to reduce the difference between the voltage of the main body and the voltage of the distal end of the endoscope. Accordingly, there is a demand for providing the endoscope with a power source voltage in the narrowest possible range.

(5) A technique disclosed in Japanese Unexamined Patent Application, First Publication No. 2011-206333 provides a function of adjusting a power source voltage based on the distal end voltage. According to the technique, the distal end voltage is monitored at all times by using a dedicated cable for determining the distal end voltage, and the power source voltage is adjusted such that the distal end voltage has an appropriate value.

SUMMARY OF THE INVENTION

(6) According to a first aspect of the present invention, an imaging device includes a camera unit and a control unit connected to each other by a power source line that transfers a first power source voltage and a video signal line that transfers a video signal. The first power source voltage transferred by the power source line is input to the camera unit as a second power source voltage. The camera unit includes an image sensor and a load circuit. The image sensor is configured to generate the video signal. The control unit includes a signal reception circuit, a voltage generation circuit, a voltage measurement circuit, and a voltage adjustment circuit. The signal reception circuit is configured to receive the video signal. The voltage generation circuit is configured to output the first power source voltage to the power source line. The voltage measurement circuit is configured to measure a value of the second power source voltage. The voltage adjustment circuit is configured to adjust a value of the first power source voltage by controlling the power source voltage generation circuit based on the value of the second power source voltage. The imaging device further includes a setting circuit configured to set the camera unit to be in a first state in a first period and set the camera unit to be in a second state in a second period different from the first period. The video signal generated by the image sensor is output to the video signal line in the first state. Output of the video signal from the image sensor to the video signal line is stopped in the second state. The load circuit is configured to consume the second power source voltage as a current in the second state. The second power source voltage is output to the video signal line in the second state. Output of the second power source voltage to the video signal line is stopped in the first state. The voltage measurement circuit is configured to measure the value of the second power source voltage transferred by the video signal line.

(7) According to a second aspect of the present invention, in the first aspect, the setting circuit may include a first switching circuit and a second switching circuit. The first switching circuit is disposed in the camera unit and is configured to connect the power source line and the video signal line to each other in the second period and disconnect the power source line and the video signal line from each other in the first period. The second switching circuit is disposed in the control unit and is configured to connect the video signal line and the signal reception circuit to each other in

the first period, disconnect the video signal line and the signal reception circuit from each other in the second period, connect the video signal line and the voltage measurement circuit to each other in the second period, and disconnect the video signal line and the voltage measurement circuit from each other in the first period.

(8) According to a third aspect of the present invention, in the second aspect, the imaging device may further include a resistance circuit including the voltage measurement circuit. The resistance circuit may have a greater resistance value than a resistance value of the video signal line. The second switching circuit may be configured to connect the video signal line and the resistance circuit to each other in the second period and disconnect the video signal line and the resistance circuit from each other in the first period.

(9) According to a fourth aspect of the present invention, in the first aspect, the camera unit may further include a transmission buffer configured to enter any one of a third state and a fourth state. When a state of the transmission buffer is the third state, the transmission buffer may be configured to output the video signal generated by the image sensor to the video signal line. When the state of the transmission buffer is the fourth state, the transmission buffer may be configured to stop output of the video signal to the video signal line. The setting circuit may be configured to set the state of the transmission buffer to the third state in the first period and set the state of the transmission buffer to the fourth state in the second period.

(10) According to a fifth aspect of the present invention, in the first aspect, the voltage adjustment circuit may be configured to calculate a resistance value of the power source line based on the value of the first power source voltage, the value of the second power source voltage, and a value of a current that flows through the power source line and adjust the value of the first power source voltage based on the resistance value.

(11) According to a sixth aspect of the present invention, an endoscope system includes both a scope that has a distal end and is to be inserted into a living body and the imaging device. The camera unit is disposed in the distal end.

(12) According to a seventh aspect of the present invention, an imaging device includes a camera unit and a control unit connected to each other by a power source line that transfers a first power source voltage, a video signal line that transfers a video signal, and a control line that transfers a control signal. The first power source voltage transferred by the power source line is input to the camera unit as a second power source voltage. The camera unit includes an image sensor and a load circuit. The image sensor is configured to generate the video signal in accordance with the control signal. The control unit includes a signal reception circuit, a control signal generation circuit, a voltage generation circuit, a voltage measurement circuit, and a voltage adjustment circuit. The signal reception circuit is configured to receive the video signal. The control signal generation circuit is configured to generate the control signal and output the generated control signal to the control signal line. The voltage generation circuit is configured to output the first power source voltage to the power source line. The voltage measurement circuit is configured to measure a value of the second power source voltage. The voltage adjustment circuit is configured to adjust a value of the first power source voltage by controlling the power source voltage generation circuit based on the value of the second power source voltage. The imaging device further includes a setting circuit configured to set the camera unit to be in a first state in a first period and set the camera unit to be in a second state in a second period different from the first period. The control signal is output to the control signal line in the first state. Output of the control signal to the control signal line is stopped in the second state. The load circuit is configured to consume the second power source voltage as a current in the second state. The second power source voltage is output to the control signal line in the second state. Output of the second power source voltage to the control signal line is stopped in the first state. The voltage measurement circuit is configured to measure the value of the second power source voltage transferred by the control signal line.

(13) According to an eighth aspect of the present invention, an imaging device includes a camera

unit and a control unit connected to each other by a power source line that transfers a first power source voltage, a video signal line that transfers a video signal, and a control line that transfers a control signal. The first power source voltage transferred by the power source line is input to the camera unit as a second power source voltage. The camera unit includes an image sensor and a load circuit. The image sensor is configured to generate the video signal in accordance with the control signal. The control unit includes a signal reception circuit, a control signal generation circuit, a voltage generation circuit, a voltage measurement circuit, and a voltage adjustment circuit. The signal reception circuit is configured to receive the video signal. The control signal generation circuit is configured to generate the control signal and output the generated control signal to the control signal line. The voltage generation circuit is configured to output the first power source voltage to the power source line. The voltage measurement circuit is configured to measure a value of the second power source voltage. The voltage adjustment circuit is configured to adjust a value of the first power source voltage by controlling the power source voltage generation circuit based on the value of the second power source voltage. Any one of the power source line, the video signal line, and the control signal line includes a first signal line and a second signal line connected in parallel to the camera unit and the control unit. The imaging device further includes a setting circuit configured to set the camera unit to be in a first state in a first period and set the camera unit to be in a second state in a second period different from the first period. Any one of the first power source voltage, the video signal, and the control signal is transferred by the first signal line and the second signal line in the first state. The load circuit is configured to consume the second power source voltage as a current in the second state. The second power source voltage is output to any one of the first signal line and the second signal line in the second state. The voltage measurement circuit is configured to measure the value of the second power source voltage transferred by any one of the first signal line and the second signal line.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram showing a configuration of an endoscope system according to a first embodiment of the present invention.
- (2) FIG. 2 is a block diagram showing a configuration of the endoscope system according to the first embodiment of the present invention.
- (3) FIG. 3 is a timing chart showing an operation of the endoscope system according to the first embodiment of the present invention.
- (4) FIG. 4 is a block diagram showing a configuration of an endoscope system according to a second embodiment of the present invention.
- (5) FIG. 5 is a block diagram showing a configuration of an endoscope system according to a third embodiment of the present invention.
- (6) FIG. 6 is a block diagram showing a configuration of an endoscope system according to a fourth embodiment of the present invention.
- (7) FIG. 7 is a block diagram showing a configuration of an endoscope system according to a fifth embodiment of the present invention.
- (8) FIG. 8 is a block diagram showing a configuration of an endoscope system according to a sixth embodiment of the present invention.
- (9) FIG. 9 is a block diagram showing a configuration of an endoscope system according to a seventh embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

- (10) Hereinafter, embodiments of the present invention will be described with reference to the drawings. Each of the embodiments will be described in detail by using an endoscope system as an

example of an imaging device.

First Embodiment

(11) FIG. 1 shows a configuration of an endoscope system 1 according to a first embodiment of the present invention. The endoscope system 1 shown in FIG. 1 includes an endoscope insertion unit 2, a transmission cable 3, an operation unit 4, a connector unit 5, a processor 6, and a display device 7. The endoscope insertion unit 2, the transmission cable 3, the operation unit 4, and the connector unit 5 constitute a scope.

(12) The endoscope insertion unit 2 includes an insertion unit 2a. The insertion unit 2a is part of the transmission cable 3. The insertion unit 2a is to be inserted inside a living body, which is a subject. The endoscope insertion unit 2 generates a video signal by imaging the inside of the subject. The endoscope insertion unit 2 outputs the generated video signal to the processor 6. A camera unit 9 shown in FIG. 2 is disposed in a distal end 2b of the insertion unit 2a. In the insertion unit 2a, the operation unit 4 is connected to the end part opposite the distal end 2b. The operation unit 4 receives various operations for the endoscope insertion unit 2 from a user.

(13) The transmission cable 3 connects the camera unit 9 and the connector unit 5. The video signal generated by the camera unit 9 is output to the connector unit 5 via the transmission cable 3.

(14) The connector unit 5 is connected to the endoscope insertion unit 2 and the processor 6. The connector unit 5 performs predetermined processing on the video signal output from the endoscope insertion unit 2. The connector unit 5 outputs the video signal to the processor 6.

(15) The processor 6 performs image processing on the video signal output from the connector unit 5. Furthermore, the processor 6 centrally controls the entire endoscope system 1.

(16) The display device 7 displays a video based on the video signal processed by the processor 6. In addition, the display device 7 displays various kinds of information related to the endoscope system 1.

(17) The endoscope system 1 includes a light source device that generates illumination light emitted to the subject. The light source device is not shown in FIG. 1.

(18) FIG. 2 shows an internal configuration of the endoscope system 1. The endoscope system 1 shown in FIG. 2 includes the camera unit 9 and the processor 6. The camera unit 9 is disposed in the distal end 2b of the scope. The operation unit 4, the connector unit 5, and the display device 7 are not shown in FIG. 2. The transmission cable 3 shown in FIG. 1 includes a power source line 30 and a video signal line 31 shown in FIG. 2.

(19) The camera unit 9 includes a power source terminal 900, a load circuit 901, a transmission buffer 902, a switch 903, and a video terminal 904. The load circuit 901 includes a switching control circuit 905 and an imager 906. At least one of the transmission buffer 902, the switch 903, and the switching control circuit 905 may be disposed in the imager 906.

(20) The processor 6 includes a voltage generation circuit 600, a voltage-and-current measurement circuit 601, a power source terminal 602, a video terminal 603, a switch 604, a switching control circuit 605, a resistor 606, a video-signal-processing circuit 607, a resistance adjustment circuit 608, a voltage measurement circuit 609, a resistance calculation circuit 610, and a voltage control circuit 611. The processor 6 is a control unit. All or part of the configuration of the processor 6 shown in FIG. 2 may be disposed in the operation unit 4 or the connector unit 5 shown in FIG. 1.

(21) A schematic configuration of the endoscope system 1 will be described. The camera unit 9 and the processor 6 are connected to each other by both the power source line 30 that transfers a first power source voltage and the video signal line 31 that transfers a video signal. The first power source voltage transferred by the power source line 30 is input to the camera unit 9 as a second power source voltage. The imager 906 generates a video signal by using the second power source voltage. The video-signal-processing circuit 607 (signal reception circuit) receives the video signal transferred by the video signal line 31. The voltage generation circuit 600 generates a first power source voltage and outputs the generated first power source voltage to the power source line 30. The voltage measurement circuit 609 measures a value of the second power source voltage. The

voltage control circuit **611** and the resistance calculation circuit **610** (voltage adjustment circuit) adjusts a value of the first power source voltage based on the value of the second power source voltage.

(22) The switch **903** and the switch **604** (setting circuit) set the camera unit **9** and the processor **6** to be in a first state in a first period and set the camera unit **9** and the processor **6** to be in a second state in a second period different from the first period. The video signal generated by the imager **906** is output to the video signal line **31** in the first state. The output of the video signal from the imager **906** to the video signal line **31** is stopped in the second state. The load circuit **901** consumes the second power source voltage as a current in the second state. The second power source voltage is output to the video signal line **31** in the second state. The output of the second power source voltage to the video signal line **31** is stopped in the first state. The voltage measurement circuit **609** measures a value of the second power source voltage transferred by the video signal line **31**.

(23) A detailed configuration of the endoscope system **1** will be described. For example, the voltage generation circuit **600** is a voltage regulator. The voltage generation circuit **600** generates the first power source voltage, which is a direct-current (DC) voltage. The voltage-and-current measurement circuit **601** measures both a value (voltage value) of the first power source voltage generated by the voltage generation circuit **600** and a value (current value) of a current (DC current) that flows through the power source line **30**. The voltage-and-current measurement circuit **601** outputs the measured voltage value and current value to the resistance calculation circuit **610**.

(24) The first power source voltage generated by the voltage generation circuit **600** is input to the power source terminal **602** via the voltage-and-current measurement circuit **601**. The power source terminal **602** is connected to the power source line **30**. The power source terminal **602** outputs the first power source voltage to the power source line **30**. The power source line **30** is a signal line disposed in the transmission cable **3**. The power source line **30** transfers the first power source voltage output from the power source terminal **602** to the camera unit **9**.

(25) The power source terminal **900** is connected to the power source line **30**. The first power source voltage transferred by the power source line **30** is input to the power source terminal **900**. The power source terminal **900** outputs the first power source voltage to each circuit in the camera unit **9** as the second power source voltage. The second power source voltage is a power source voltage transferred by the power source line **30** to the camera unit **9** and is a voltage on a path including a path from the power source terminal **900** to the imager **906**. A voltage drop is generated due to the DC resistance of the power source line **30**, and the second power source voltage is attenuated. Therefore, the value of the second power source voltage is less than that of the first power source voltage in the processor **6**.

(26) The load circuit **901** is connected to the power source terminal **900**. The load circuit **901** has a DC resistance value and consumes the second power source voltage as a current. The imager **906** is an image sensor such as a complementary metal-oxide semiconductor (CMOS) sensor. The imager **906** includes a plurality of pixels and generates a video signal having a voltage generated based on the second power source voltage.

(27) In the example shown in FIG. 2, the load circuit **901** includes the imager **906**. The load circuit **901** does not need to include the imager **906**. The load circuit **901** may include only the imager **906**.

(28) For example, the transmission buffer **902** is a transistor including a gate terminal, a source terminal, and a drain terminal and constitutes a source follower. The gate terminal of the transmission buffer **902** is connected to the imager **906**. One of the source terminal and the drain terminal of the transmission buffer **902** is connected to the power source terminal **900**, and the other of the source terminal and the drain terminal of the transmission buffer **902** is connected to the video terminal **904**.

(29) The switch **903** (first switching circuit) constitutes a setting circuit. The switch **903** is connected to the power source terminal **900** and the video terminal **904**. The switch **903** connects

the power source line **30** and the video signal line **31** to each other in the second period and disconnects the power source line **30** and the video signal line **31** from each other in the first period. For example, the first period is a period during which the imager **906** outputs the video signal. The second period is all or part of a period excluding the first period. For example, the second period is a blanking period. The blanking period is at least one of a horizontal blanking period and a vertical blanking period.

(30) The switch **903** enters any one of an ON state and an OFF state. The switch **903** can switch between the ON state and the OFF state. When the state of the switch **903** is the ON state, the switch **903** connects the power source line **30** and the video signal line **31** to each other. When the state of the switch **903** is the OFF state, the switch **903** disconnects the power source line **30** and the video signal line **31** from each other.

(31) The switching control circuit **905** outputs a switching control signal to the switch **903**, thus controlling the state of the switch **903**. The switching control circuit **905** sets the state of the switch **903** to the OFF state in the first period. At this time, the transmission buffer **902** outputs the video signal to the video terminal **904**. The switching control circuit **905** sets the state of the switch **903** to the ON state in the second period. At this time, the power source line **30** and the video signal line **31** are short-circuited, and the second power source voltage is output to the video terminal **904**. The voltages of the source terminal and the drain terminal of the transmission buffer **902** become almost the same, and the transmission buffer **902** stops the output of the video signal.

(32) The video terminal **904** is connected to the video signal line **31**. The video terminal **904** outputs the video signal or the second power source voltage to the video signal line **31**. The video signal line **31** is a signal line disposed in the transmission cable **3**. The video signal line **31** transfers the video signal output from the transmission buffer **902** to the processor **6** in the first period. The video signal line **31** transfers the second power source voltage output from the power source terminal **900** to the processor **6** in the second period.

(33) The video terminal **603** is connected to the video signal line **31**. The video signal or the second power source voltage transferred by the video signal line **31** is input to the video terminal **603**.

(34) The switch **604** (second switching circuit) constitutes a setting circuit. The switch **604** is connected to the video terminal **603**, the voltage measurement circuit **609**, and the video-signal-processing circuit **607**. The switch **604** connects the video signal line **31** and the video-signal-processing circuit **607** to each other in the first period and disconnects the video signal line **31** and the video-signal-processing circuit **607** from each other in the second period. The switch **604** connects the video signal line **31** and the voltage measurement circuit **609** to each other in the second period and disconnects the video signal line **31** and the voltage measurement circuit **609** from each other in the first period.

(35) The switch **604** enters any one of a video output state and a power source output state. The switch **604** can switch between the video output state and the power source output state. When the state of the switch **604** is the video output state, the switch **604** connects the video signal line **31** and the video-signal-processing circuit **607** to each other and disconnects the video signal line **31** and the voltage measurement circuit **609** from each other. When the state of the switch **604** is the power source output state, the switch **604** disconnects the video signal line **31** and the video-signal-processing circuit **607** from each other and connects the video signal line **31** and the voltage measurement circuit **609** to each other.

(36) The switching control circuit **605** outputs a switching control signal to the switch **604**, thus controlling the state of the switch **604**. The switching control circuit **605** sets the state of the switch **604** to the video output state in the first period. At this time, the video signal transferred by the video signal line **31** and input to the video terminal **603** is output to the video-signal-processing circuit **607** via the switch **604**. The switching control circuit **605** sets the state of the switch **604** to the power source output state in the second period. At this time, the second power source voltage transferred by the video signal line **31** and input to the video terminal **603** is output to the voltage

measurement circuit **609** via the switch **604**.

(37) The resistor **606** is connected to the switch **604** and the video-signal-processing circuit **607**. The resistor **606** is a terminal resistor. The video signal is input to the video-signal-processing circuit **607** in the first period. For example, the video-signal-processing circuit **607** is an analog front end (AFE). The video-signal-processing circuit **607** receives the video signal and performs predetermined signal processing on the video signal. In a case in which the resistor **606** is unnecessary or has a high resistance value, the switch **604** and the switching control circuit **605** are not necessarily disposed.

(38) The resistance adjustment circuit **608** and the voltage measurement circuit **609** are connected to the switch **604**. The resistance adjustment circuit **608** and the voltage measurement circuit **609** constitute a resistance circuit having a DC resistance value. The resistance value of the input terminal (the input terminal of the voltage measurement circuit **609**) of the resistance circuit is extremely high, and the resistance circuit is in a high-impedance state (Hi-Z). The second power source voltage is input to the input terminal of the resistance circuit in the second period.

(39) Since the resistance circuit is in the high-impedance state, a current hardly flows through the video signal line **31** in the second period and the value of the second power source voltage input to the resistance circuit is almost the same as that of the second power source voltage input to the power source terminal **900** in the camera unit **9**.

(40) The resistance circuit needs to have at least a greater resistance value than that of the video signal line **31**. The resistance value of the video signal line **31** and an approximate resistance value of the load circuit **901** are known. The switch **604** connects the video signal line **31** and the resistance circuit to each other in the second period and disconnects the video signal line **31** and the resistance circuit from each other in the first period.

(41) For example, the resistance adjustment circuit **608** is an input impedance of the voltage measurement circuit **609** or a capacitance element through which a DC current does not flow.

(42) The voltage measurement circuit **609** measures a value of the second power source voltage in the second period. The voltage measurement circuit **609** outputs the measured value of the second power source voltage to the resistance calculation circuit **610**. The video-signal-processing circuit **607** may also have the function of the voltage measurement circuit **609**.

(43) The resistance calculation circuit **610** and the voltage control circuit **611** constitute a voltage adjustment circuit. The resistance calculation circuit **610** calculates a resistance value (DC resistance value) of the power source line **30** based on the value of the first power source voltage, the value of the second power source voltage, and the value of the current that flows through the power source line **30**. These values meet a condition shown in the following Expression (1).

$$VDD - V_{cis} = R_{cable} \cdot I_{vdd} \quad (1)$$

(44) In Expression (1), a value VDD indicates the value of the first power source voltage, a value V_{cis} indicates the value of the second power source voltage, a resistance value R_{cable} indicates the resistance value of the power source line **30**, and a current value I_{vdd} indicates the value of the current that flows through the power source line **30**. Accordingly, the resistance calculation circuit **610** can calculate the resistance value R_{cable} in accordance with the following Expression (2).

$$R_{cable} = (VDD - V_{cis}) / I_{vdd} \quad (2)$$

(45) The resistance calculation circuit **610** calculates a control value of the first power source voltage by using the calculated resistance value. The resistance calculation circuit **610** outputs the calculated control value to the voltage control circuit **611**. The voltage control circuit **611** controls the voltage generation circuit **600** based on the control value, thus adjusting a value of the first power source voltage to be generated by the voltage generation circuit **600**. For example, when the resistance value R_{cable} decreases, the amount of the voltage drop in the power source line **30** decreases and the value of the second power source voltage increases. Therefore, the voltage control circuit **611** reduces the value of the first power source voltage. When the resistance value R_{cable} increases, the amount of the voltage drop in the power source line **30** increases and the

value of the second power source voltage decreases. Therefore, the voltage control circuit **611** increases the value of the first power source voltage. In addition, when the current value measured by the voltage-and-current measurement circuit **601** increases, the amount of the voltage drop in the power source line **30** increases and the value of the second power source voltage decreases. Therefore, the voltage control circuit **611** increases the value of the first power source voltage. When the current value measured by the voltage-and-current measurement circuit **601** decreases, the amount of the voltage drop in the power source line **30** decreases and the value of the second power source voltage increases. Therefore, the voltage control circuit **611** reduces the value of the first power source voltage.

(46) The voltage control circuit **611** adjusts the value of the first power source voltage such that the value of the second power source voltage input to the power source terminal **900** matches a recommended voltage value for the operation of the imager **906**. For example, the recommended voltage value is 3.3 V. The value VDD of the first power source voltage, the value Vcis of the second power source voltage, the current value Ivdd, and the resistance value Rcable meet the condition shown in Expression (1) described above. In order for the value Vcis of the second power source voltage to be 3.3 V, a condition shown in the following Expression (3) needs to be met. The voltage control circuit **611** controls the voltage generation circuit **600** such that the value of the first power source voltage to be generated by the voltage generation circuit **600** matches a value VDD' shown in Expression (3).

$$VDD' = 3.3 + Rcable * Ivdd \quad (3)$$

(47) In addition, by considering Expression (1) and Expression (3), the value VDD' is also expressed by Expression (4). The resistance calculation circuit **610** and the voltage control circuit **611** may control the voltage generation circuit **600** in accordance with Expression (4).

$$VDD' = 3.3 + VDD - Vcis \quad (4)$$

(48) FIG. 3 shows an operation of the endoscope system **1**. In FIG. 3, an operation phase of the imager **906** is shown. In addition, waveforms of the switching control signal input to the switch **903** or the switch **604**, the second power source voltage output from the power source terminal **900**, the signal input to the video terminal **603**, and the current that flows through the power source line **30** are shown in FIG. 3. The horizontal direction in FIG. 3 indicates time, and the vertical direction in FIG. 3 indicates a voltage value or a current value.

(49) The operation phase of the imager **906** repeats reading of the video signal and a blanking period. The blanking period periodically occurs. While the imager **906** outputs the video signal in the first period, the voltage of the switching control signal is a low level. At this time, the state of the switch **903** is the OFF state, and the state of the switch **604** is the video output state. Therefore, the video signal line **31** transfers the video signal, and the video-signal-processing circuit **607** receives the video signal.

(50) When the blanking period is started, the imager **906** stops the output of the video signal. The voltage of the switching control signal changes to a high level in the blanking period. At this time, the first period is completed, and the second period is started. The state of the switch **903** changes to the ON state, and the state of the switch **604** changes to the power source output state. Therefore, the video signal line **31** transfers the second power source voltage, and the second power source voltage is input to the voltage measurement circuit **609**. The voltage measurement circuit **609** measures a value of the second power source voltage.

(51) The video signal passes through the video signal line **31** in the first period, and a current flows through the video signal line **31**. On the other hand, the current hardly flows through the video signal line **31** in the second period. The value of the current that flows through the video signal line **31** in the second period is less than that in the first period. Therefore, the value of the current that flows through the power source line **30** in the second period is less than that in the first period. If the current that flows through the power source line **30** is reduced, the amount of the voltage drop in the power source line **30** is reduced. Therefore, the value of the second power source voltage in

the second period is greater than that in the first period.

(52) The voltage of the switching control signal changes to the low level before the blanking period is completed. At this time, the second period is completed, and the first period is started. The state of the switch **903** changes to the OFF state, and the state of the switch **604** changes to the video output state. Therefore, the output of the second power source voltage to the video signal line **31** is stopped.

(53) When the blanking period is completed, the imager **906** starts the output of the video signal. The video signal line **31** transfers the video signal, and the video-signal-processing circuit **607** receives the video signal.

(54) In one or more of two or more blanking periods that periodically occur, the voltage measurement circuit **609** calculates a value of the second power source voltage and the resistance calculation circuit **610** calculates a resistance value of the power source line **30**. Each time a blanking period occurs, the voltage measurement circuit **609** does not need to calculate a value of the second power source voltage and the resistance calculation circuit **610** does not need to calculate a resistance value of the power source line **30**.

(55) In the first embodiment, the endoscope system **1** can monitor a power source voltage (second power source voltage) provided to the imager **906**. The video signal line **31** is used for transferring the video signal and the second power source voltage, and the endoscope system **1** does not need to monitor the second power source voltage at all times. Therefore, a dedicated cable for transferring the second power source voltage is unnecessary, and the miniaturization of the camera unit **9** is not prevented.

(56) The endoscope system **1** can calculate the amount of a voltage drop in the power source line **30** by calculating the resistance value of the power source line **30**. The endoscope system **1** can directly adjust the value of the first power source voltage to be generated by the voltage generation circuit **600** by controlling the voltage generation circuit **600** based on the amount.

(57) A switch through which a large current flows has a large area. Since a current hardly flows through the switch **903** in the second period, the switch **903** does not need to have a large area. In addition, since a switch capable of switching between the ON state and the OFF state is used as the switch **903**, the switch **903** can be miniaturized.

Second Embodiment

(58) FIG. **4** shows an internal configuration of an endoscope system **1a** according to a second embodiment of the present invention. The same configuration as that shown in FIG. **2** will not be described. The endoscope system **1a** shown in FIG. **4** includes a camera unit **9a** and a processor **6**.

(59) The camera unit **9a** includes a power source terminal **900**, a load circuit **901**, a transmission buffer **902**, a switch **903**, a video terminal **904**, and a switch **910**. The load circuit **901** includes a switching control circuit **905** and an imager **906**. At least one of the transmission buffer **902**, the switch **903**, and the switch **910** may be disposed in the imager **906**. The processor **6** is the same as that shown in FIG. **2**.

(60) The transmission buffer **902** enters any one of a video output state (third state) and an output stoppage state (fourth state). When the state of the transmission buffer **902** is the video output state, the transmission buffer **902** outputs the video signal generated by the imager **906** to the video signal line **31**. When the state of the transmission buffer **902** is the output stoppage state, the transmission buffer **902** stops the output of the video signal to the video signal line **31**. The switch **910** (setting circuit) sets the state of the transmission buffer **902** to the video output state in the first period, and sets the state of the transmission buffer **902** to the output stoppage state in the second period.

(61) The switch **910** is connected to the imager **906** and the transmission buffer **902**. A gate terminal of the transmission buffer **902** is connected to the switch **910**. The switch **910** connects the imager **906** and the transmission buffer **902** to each other in the first period and disconnects the imager **906** and the transmission buffer **902** from each other in the second period.

- (62) The switch **910** enters any one of the video output state and the output stoppage state. The switch **910** can switch between the video output state and the output stoppage state. When the state of the switch **910** is the video output state, the switch **910** connects the imager **906** and the transmission buffer **902** to each other. When the state of the switch **910** is the output stoppage state, the switch **910** disconnects the imager **906** and the transmission buffer **902** from each other.
- (63) The video signal is input to the switch **910** in the first period, and the switch **910** outputs the video signal to the transmission buffer **902**. The transmission buffer **902** outputs the video signal to the video terminal **904**.
- (64) A stoppage signal is input to the switch **910** in the second period, and the switch **910** outputs the stoppage signal to the transmission buffer **902**. The stoppage signal has a fixed voltage. For example, the voltage of the stoppage signal is a ground voltage. The voltage of the stoppage signal may be the second power source voltage. The voltage of the stoppage signal may be a voltage between the ground voltage and the second power source voltage. Since the stoppage signal is input to the gate terminal of the transmission buffer **902**, the voltage output from the transmission buffer **902** to the video terminal **904** does not change.
- (65) In the second embodiment, the endoscope system **1a** can monitor a power source voltage provided to the imager **906** and does not prevent the miniaturization of the camera unit **9a** as with the endoscope system **1** according to the first embodiment. In addition, since the transmission buffer **902** outputs a fixed voltage to the video terminal **904** in the second period, the endoscope system **1a** can reduce the amount of noise mixed into the second power source voltage output from the video terminal **904** to the video signal line **31**.

Third Embodiment

- (66) FIG. 5 shows an internal configuration of an endoscope system **1b** according to a third embodiment of the present invention. The same configuration as that shown in FIG. 2 will not be described. The endoscope system **1b** shown in FIG. 5 includes a camera unit **9b** and a processor **6b**. The transmission cable **3** shown in FIG. 1 includes a power source line **30**, a video signal line **31**, and a control signal line **32** shown in FIG. 5. The camera unit **9b** and the processor **6b** are connected to each other by the power source line **30**, the video signal line **31**, and the control signal line **32**.
- (67) The camera unit **9b** includes a power source terminal **900**, a load circuit **901b**, a transmission buffer **902**, a video terminal **904**, a switch **920**, a switch **921**, a control terminal **922**, and a load circuit **923**. The load circuit **901b** includes an imager **906**. At least one of the transmission buffer **902**, the switch **920**, the switch **921**, and the load circuit **923** may be disposed in the imager **906**.
- (68) The processor **6b** includes a voltage generation circuit **600**, a voltage-and-current measurement circuit **601**, a power source terminal **602**, a video terminal **603**, a switch **604**, a resistor **606**, a video-signal-processing circuit **607**, a resistance adjustment circuit **608**, a voltage measurement circuit **609**, a resistance calculation circuit **610**, a voltage control circuit **611**, a timing generator (TG) **620**, and a control terminal **621**. All or part of the configuration of the processor **6b** shown in HG. 5 may be disposed in the operation unit **4** or the connector unit **5** shown in FIG. 1.
- (69) The TG **620** generates a switching control signal used for controlling the state of each of the switches **604**, **920**, and **921**. The TG **620** outputs the generated switching control signal to the switch **604** and the control terminal **621**.
- (70) The state of the switch **604** is controlled based on the switching control signal generated by the TG **620**. The state of the switch **604** is set to the video output state in the first period. At this time, the video signal transferred by the video signal line **31** and input to the video terminal **603** is output to the video-signal-processing circuit **607** via the switch **604**. The state of the switch **604** is set to the power source output state in the second period. At this time, the second power source voltage transferred by the video signal line **31** and input to the video terminal **603** is output to the voltage measurement circuit **609** via the switch **604**.
- (71) The switching control signal generated by the TG **620** is input to the control terminal **621**. The

control terminal **621** is connected to the control signal line **32**. The control terminal **621** outputs the switching control signal to the control signal line **32**. The control signal line **32** is a signal line disposed in the transmission cable **3**. The control signal line **32** transfers the switching control signal output from the control terminal **621** to the camera unit **9b**.

(72) The control terminal **922** is connected to the control signal line **32**. The switching control signal transferred by the control signal line **32** is input to the control terminal **922**. The control terminal **922** outputs the switching control signal to the switch **920** and the switch **921**.

(73) The switch **920** and the switch **921** (first switching circuit) constitute a setting circuit. The switch **920** is connected to the power source terminal **900**, the load circuit **901b**, the switch **921**, and the load circuit **923**. The switch **921** is connected to the video terminal **904**, the transmission buffer **902**, and the switch **920**.

(74) The switch **920** and the switch **921** connect the power source line **30** and the video signal line **31** to each other in the second period and disconnect the power source line **30** and the video signal line **31** from each other in the first period. The switch **921** connects the transmission buffer **902** and the video signal line **31** to each other in the first period and disconnects the transmission buffer **902** and the video signal line **31** from each other in the second period. The switch **920** and the switch **921** connect the power source line **30** and the load circuit **923** to each other in the second period and disconnects the power source line **30** and the load circuit **923** from each other in the first period.

(75) The switch **920** enters any one of an imager-driving state and a power source output state. The switch **920** can switch between the imager-driving state and the power source output state. When the state of the switch **920** is the imager-driving state, the switch **920** connects the power source line **30** and the load circuit **901b** to each other, disconnects the power source line **30** and the switch **921** from each other, and disconnects the power source line **30** and the load circuit **923** from each other. When the state of the switch **920** is the power source output state, the switch **920** disconnects the power source line **30** and the load circuit **901b** from each other, connects the power source line **30** and the switch **921** to each other, and connects the power source line **30** and the load circuit **923** to each other.

(76) The state of the switch **920** is controlled based on the switching control signal output from the control terminal **922**. The state of the switch **920** is set to the imager-driving state in the first period. At this time, the second power source voltage transferred by the power source line **30** and input to the power source terminal **900** is output to the imager **906** via the switch **920**, and the imager **906** outputs the video signal. The state of the switch **920** is set to the power source output state in the second period. At this time, the second power source voltage transferred by the power source line **30** and input to the power source terminal **900** is output to the switch **921** and the load circuit **923** via the switch **920**.

(77) The load circuit **923** has a DC resistance value and consumes the second power source voltage as a current. For example, the load circuit **923** is a resistance element, a current source, or the like. Although a current flows through the load circuit **923**, a current hardly flows through the video signal line **31**.

(78) The switch **921** enters any one of a video output state and a power source output state. The switch **921** can switch between the video output state and the power source output state. When the state of the switch **921** is the video output state, the switch **921** connects the transmission buffer **902** and the video signal line **31** to each other and disconnects the switch **920** and the video signal line **31** from each other. When the state of the switch **921** is the power source output state, the switch **921** disconnects the transmission buffer **902** and the video signal line **31** from each other and connects the switch **920** and the video signal line **31** to each other.

(79) The state of the switch **921** is controlled based on the switching control signal output from the control terminal **922**. The state of the switch **921** is set to the video output state in the first period. At this time, the transmission buffer **902** outputs the video signal to the video terminal **904**. The

state of the switch **921** is set to the power source output state in the second period. Since the state of each of the switches **920** and **921** is the power source output state in the second period, the power source line **30** and the video signal line **31** are short-circuited and the second power source voltage is output to the video terminal **904**. Since the transmission buffer **902** and the video signal line **31** are disconnected from each other in the second period, the transmission buffer **902** stops the output of the video signal.

(80) A switch control method in the third embodiment may be applied to the endoscope system **1** of the first embodiment or the endoscope system **1a** of the second embodiment. In other words, in the endoscope system **1** or the endoscope system **1a**, the control signal line **32** may be disposed, the switch **920** and the switch **921** may be disposed instead of the switch **903** and the switching control circuit **905**, and the TG **620** may be disposed instead of the switching control circuit **605**.

(81) In the third embodiment, the endoscope system **1b** can monitor a power source voltage provided to the imager **906** as with the endoscope system **1** according to the first embodiment.

(82) Since the resistance value of the load circuit **923** is stable, a stable second power source voltage is output to the video signal line **31**. Since the switching control circuit **605** (see FIG. 2) that generates the switching control signal is unnecessary, an increase of the circuit scale of the camera unit **9b** is restricted.

Fourth Embodiment

(83) FIG. 6 shows an internal configuration of an endoscope system **1c** according to a fourth embodiment of the present invention. The same configuration as that shown in FIG. 2 will not be described. The endoscope system **1c** shown in FIG. 6 includes a camera unit **9c** and a processor **6c**. The transmission cable **3** shown in FIG. 1 includes a power source line **30**, a video signal line **31**, and a control signal line **32** shown in FIG. 6. The camera unit **9c** and the processor **6c** are connected to each other by the power source line **30**, the video signal line **31**, and the control signal line **32**.

(84) The camera unit **9c** includes a power source terminal **900**, a load circuit **901**, a video terminal **904**, a control terminal **922**, and a switch **930**. The load circuit **901** includes a switching control circuit **905** and an imager **906**. At least one of the switching control circuit **905** and the switch **930** may be disposed in the imager **906**.

(85) The processor **6c** includes a voltage generation circuit **600**, a voltage-and-current measurement circuit **601**, a power source terminal **602**, a video terminal **603**, a switching control circuit **605**, a video-signal-processing circuit **607**, a resistance adjustment circuit **608**, a voltage measurement circuit **609**, a resistance calculation circuit **610**, a voltage control circuit **611**, a timing generator (TG) **620**, a control terminal **621**, and a switch **630**. All or part of the configuration of the processor **6c** shown in FIG. 6 may be disposed in the operation unit **4** or the connector unit **5** shown in FIG. 1.

(86) The TG **620** (control signal generation circuit) generates an imager control signal used for controlling the state of the imager **906**. For example, the imager control signal is a clock signal, a synchronization signal, a data signal, or the like. For example, the data signal is used for rewriting a value of a register that stores the state of the switching control circuit **905** of the imager **906**. The TG **620** outputs the generated imager control signal to the switch **630**.

(87) The switch **630** (second switching circuit) constitutes a setting circuit. The switch **630** is connected to the control terminal **621**, the voltage measurement circuit **609**, and the TG **620**. The switch **630** connects the control signal line **32** and the TG **620** to each other in the first period and disconnects the control signal line **32** and the TG **620** from each other in the second period. The switch **630** connects the control signal line **32** and the voltage measurement circuit **609** to each other in the second period and disconnects the control signal line **32** and the voltage measurement circuit **609** from each other in the first period.

(88) The switch **630** enters any one of a control signal output state and a power source output state. The switch **630** can switch between the control signal output state and the power source output

state. When the state of the switch **630** is the control signal output state, the switch **630** connects the control signal line **32** and the TG **620** to each other and disconnects the control signal line **32** and the voltage measurement circuit **609** from each other. When the state of the switch **630** is the power source output state, the switch **630** disconnects the control signal line **32** and the TG **620** from each other and connects the control signal line **32** and the voltage measurement circuit **609** to each other.

(89) The switching control circuit **605** outputs a switching control signal to the switch **630**, thus controlling the state of the switch **630**. The switching control circuit **605** sets the state of the switch **630** to the control signal output state in the first period. At this time, the imager control signal output from the TG **620** is output to the control signal line **32** via the switch **630** and the control terminal **621**. The switching control circuit **605** sets the state of the switch **630** to the power source output state in the second period. As described later, the control signal line **32** transfers the second power source voltage from the camera unit **9c** to the processor **6c** in the second period. The second power source voltage transferred by the control signal line **32** and input to the control terminal **621** is output to the voltage measurement circuit **609** via the switch **630**.

(90) The imager control signal generated by the TG **620** is input to the control terminal **621**. The control terminal **621** is connected to the control signal line **32**. The control terminal **621** outputs the imager control signal to the control signal line **32**. The control signal line **32** transfers the imager control signal output from the control terminal **621** to the camera unit **9c**.

(91) The control terminal **922** is connected to the control signal line **32**. The imager control signal transferred by the control signal line **32** is input to the control terminal **922**. The control terminal **922** outputs the imager control signal to the switch **930**.

(92) The switch **930** (first switching circuit) constitutes a setting circuit. The switch **930** is connected to the control terminal **922**, the power source terminal **900**, and the imager **906**. The switch **930** connects the power source line **30** and the control signal line **32** to each other in the second period and disconnects the power source line **30** and the control signal line **32** from each other in the first period. The switch **930** connects the imager **906** and the control signal line **32** to each other in the first period and disconnects the imager **906** and the control signal line **32** from each other in the second period.

(93) The switch **930** enters any one of an imager control state and a power source output state. The switch **930** can switch between the imager control state and the power source output state. When the state of the switch **930** is the imager control state, the switch **930** connects the imager **906** and the control signal line **32** to each other and disconnects the power source line **30** and the control signal line **32** from each other. When the state of the switch **930** is the power source output state, the switch **930** disconnects the imager **906** and the control signal line **32** from each other and connects the power source line **30** and the control signal line **32** to each other.

(94) The switching control circuit **905** outputs a switching control signal to the switch **930**, thus controlling the state of the switch **930**. The switching control circuit **905** sets the state of the switch **930** to the imager control state in the first period. At this time, the imager control signal transferred by the control signal line **32** and input to the control terminal **922** is output to the imager **906** via the switch **930**. The imager **906** generates the video signal in accordance with the imager control signal by using the second power source voltage. The switching control circuit **905** sets the state of the switch **930** to the power source output state in the second period. At this time, the power source line **30** and the control signal line **32** are short-circuited, and the second power source voltage is output to the control terminal **922**.

(95) The control terminal **922** outputs the second power source voltage to the control signal line **32**. The control signal line **32** transfers the second power source voltage output from the control terminal **922** to the processor **6c**. The second power source voltage transferred by the control signal line **32** is input to the control terminal **621**. The control terminal **621** outputs the second power source voltage to the voltage measurement circuit **609** via the switch **630**.

(96) The setting circuit including the switch **630** and the switch **930** sets the camera unit **9c** and the processor **6c** to be in a first state in the first period and sets the camera unit **9c** and the processor **6c** to be in a second state in the second period. The imager control signal is output to the control signal line **32** in the first state. The output of the imager control signal to the control signal line **32** is stopped in the second state. The load circuit **901** consumes the second power source voltage as a current in the second state. The second power source voltage is output to the control signal line **32** in the second state. The output of the second power source voltage to the control signal line **32** is stopped in the first state. The voltage measurement circuit **609** measures a value of the second power source voltage transferred by the control signal line **32**.

(97) The imager control signal is not provided to the imager **906** in the second period. The imager **906** can operate in accordance with the clock signal in the camera unit **9c** in the second period.

(98) In the fourth embodiment, the endoscope system **1c** can monitor a power source voltage provided to the imager **906** as with the endoscope system **1** according to the first embodiment. The control signal line **32** is used for transferring the imager control signal and the second power source voltage, and the endoscope system **1c** does not need to monitor the second power source voltage at all times. Therefore, a dedicated cable for transferring the second power source voltage is unnecessary, and the miniaturization of the camera unit **9c** is not prevented.

(99) The video signal line **31** is not used for transferring the second power source voltage. Therefore, the imager **906** can output the video signal also in the second period.

Fifth Embodiment

(100) FIG. **7** shows an internal configuration of an endoscope system **1d** according to a fifth embodiment of the present invention. The same configuration as that shown in FIG. **2** will not be described. The endoscope system **1d** shown in FIG. **7** includes a camera unit **9d** and a processor **6d**. The transmission cable **3** shown in FIG. **1** includes a power source line **30**, a video signal line **31**, and a power source line **33** shown in FIG. **7**. The camera unit **9d** and the processor **6d** are connected to each other by the power source line **30**, the video signal line **31**, and the power source line **33**.

(101) The camera unit **9d** includes a power source terminal **900**, a load circuit **901d**, a video terminal **904**, and a power source terminal **940**. The load circuit **901d** includes an imager **906**.

(102) The processor **6d** includes a voltage generation circuit **600**, a voltage-and-current measurement circuit **601**, a power source terminal **602**, a video terminal **603**, a switching control circuit **605**, a video-signal-processing circuit **607**, a resistance adjustment circuit **608**, a voltage measurement circuit **609**, a resistance calculation circuit **610**, a voltage control circuit **611**, a switch **640**, and a power source terminal **641**. All or part of the configuration of the processor **6d** shown in FIG. **7** may be disposed in the operation unit **4** or the connector unit **5** shown in FIG. **1**.

(103) The switch **640** (second switching circuit) constitutes a setting circuit. The switch **640** is connected to the power source terminal **641**, the voltage-and-current measurement circuit **601**, and the voltage measurement circuit **609**. The switch **640** connects the power source line **33** and the voltage-and-current measurement circuit **601** to each other in the first period and disconnects the power source line **33** and the voltage-and-current measurement circuit **601** from each other in the second period. The switch **640** connects the power source line **33** and the voltage measurement circuit **609** to each other in the second period and disconnects the power source line **33** and the voltage measurement circuit **609** from each other in the first period.

(104) The switch **640** enters any one of a first power source output state and a second power source output state. The switch **640** can switch between the first power source output state and the second power source output state. When the state of the switch **640** is the first power source output state, the switch **640** connects the power source line **33** and the voltage-and-current measurement circuit **601** to each other and disconnects the power source line **33** and the voltage measurement circuit **609** from each other. When the state of the switch **640** is the second power source output state, the switch **640** disconnects the power source line **33** and the voltage-and-current measurement circuit

601 from each other and connects the power source line **33** and the voltage measurement circuit **609** to each other.

(105) The switching control circuit **605** outputs a switching control signal to the switch **640**, thus controlling the state of the switch **640**. The switching control circuit **605** sets the state of the switch **640** to the first power source output state in the first period. At this time, the first power source voltage generated by the voltage generation circuit **600** is output to the power source line **30** via the power source terminal **602** and is output to the power source line **33** via the switch **640** and the power source terminal **641**. The switching control circuit **605** sets the state of the switch **640** to the second power source output state in the second period. As described later, the power source line **33** transfers the second power source voltage from the camera unit **9d** to the processor **6d** in the second period. The second power source voltage transferred by the power source line **33** and input to the power source terminal **641** is output to the voltage measurement circuit **609** via the switch **640**.

(106) The state of the switch **640** is set to the first power source output state in the first period. The first power source voltage generated by the voltage generation circuit **600** is input to the power source terminal **641** via the voltage-and-current measurement circuit **601** and the switch **640** in the first period. The power source terminal **641** is connected to the power source line **33**. The power source terminal **641** outputs the first power source voltage to the power source line **33**. The power source line **33** is a signal line disposed in the transmission cable **3**. For example, the sum of the cross-sectional area of the power source line **30** and the cross-sectional area of the power source line **33** is the same as the cross-sectional area of the power source line **30** shown in FIG. **2**. The power source line **33** transfers the first power source voltage output from the power source terminal **641** to the camera unit **9d**.

(107) The power source terminal **940** is connected to the power source line **33**. The first power source voltage transferred by the power source line **33** is input to the power source terminal **940**. The power source terminal **940** outputs the first power source voltage to each circuit in the camera unit **9d** as the second power source voltage. The first power source voltage is transferred to the camera unit **9d** by both the power source line **30** and the power source line **33** in the first period.

(108) The state of the switch **640** is set to the second power source output state in the second period. The power source line **33** and the voltage-and-current measurement circuit **601** are disconnected from each other, and the first power source voltage is transferred to the camera unit **9d** only by the power source line **30**.

(109) The first power source voltage transferred by the power source line **30** is input to the power source terminal **900**. The power source terminal **900** outputs the first power source voltage to the load circuit **901d** and the power source terminal **940** as the second power source voltage. The power source terminal **940** outputs the second power source voltage to the power source line **33**. The power source line **33** transfers the second power source voltage output from the power source terminal **940** to the processor **6d**.

(110) The second power source voltage transferred by the power source line **33** is input to the power source terminal **641**. Since the power source line **33** and the voltage-and-current measurement circuit **601** are disconnected from each other and the power source line **33** and the voltage measurement circuit **609** are connected to each other, the power source terminal **641** outputs the second power source voltage to the voltage measurement circuit **609** via the switch **640**.

(111) The power source line **30** (first signal line) and the power source line **33** (second signal line) are connected in parallel to the camera unit **9d** and the processor **6d**. A setting circuit including the switch **640** sets the camera unit **9d** and the processor **6d** to be in the first state in the first period and sets the camera unit **9d** and the processor **6d** to be in the second state in the second period. The first power source voltage is transferred to the camera unit **9d** by the power source line **30** and the power source line **33** in the first state. The load circuit **901d** consumes the second power source voltage as a current in the second state. The second power source voltage is output to the power

source line **33** in the second state. The voltage measurement circuit **609** measures a value of the second power source voltage transferred by the power source line **33**.

(112) In the fifth embodiment, the endoscope system **1d** can monitor a power source voltage provided to the imager **906** as with the endoscope system **1** according to the first embodiment.

(113) Since the switching control circuit **905** (see FIG. **2**) that generates the switching control signal is unnecessary, an increase of the circuit scale of the camera unit **9d** is restricted. The video signal line **31** is not used for transferring the second power source voltage. Therefore, the imager **906** can output the video signal also in the second period.

(114) The first power source voltage is transferred to the camera unit **9d** by the power source line **30** and the power source line **33** in the first period and is transferred to the camera unit **9d** only by the power source line **30** in the second period. Since the resistance value of the power source line **30** is greater than a combined resistance value of the power source line **30** and the power source line **33**, there is a possibility that the second power source voltage is likely to change due to the influence of the operation of the load circuit **901d**. Therefore, there is a possibility that the characteristics of the video signal in the second period are different from those of the video signal in the first period. However, the endoscope system **1d** can transfer the second power source voltage from the camera unit **9d** to the processor **6d** in the second period by allocating a blanking period to the second period or by allocating a period during which the imager **906** outputs a signal not required to be highly accurate to the second period.

Sixth Embodiment

(115) FIG. **8** shows an internal configuration of an endoscope system **1e** according to a sixth embodiment of the present invention. The same configuration as that shown in FIG. **2** will not be described. The endoscope system **1e** shown in FIG. **8** includes a camera unit **9e** and a processor **6e**. The transmission cable **3** shown in FIG. **1** includes a power source line **30**, a video signal line **31**, and a video signal line **34** shown in FIG. **8**. The camera unit **9e** and the processor **6e** are connected to each other by the power source line **30**, the video signal line **31**, and the video signal line **34**.

(116) The camera unit **9e** includes a power source terminal **900**, a load circuit **901**, a video terminal **904**, a video terminal **950**, and a switch **951**. The load circuit **901** includes a switching control circuit **905** and an imager **906**. At least one of the switching control circuit **905** and the switch **951** may be disposed in the imager **906**.

(117) The processor **6e** includes a voltage generation circuit **600**, a voltage-and-current measurement circuit **601**, a power source terminal **602**, a video terminal **603**, a switching control circuit **605**, a video-signal-processing circuit **607**, a resistance adjustment circuit **608**, a voltage measurement circuit **609**, a resistance calculation circuit **610**, a voltage control circuit **611**, a switch **650**, and a video terminal **651**. All or part of the configuration of the processor **6e** shown in FIG. **8** may be disposed in the operation unit **4** or the connector unit **5** shown in FIG. **1**.

(118) The switch **951** (first switching circuit) constitutes a setting circuit. The switch **951** is connected to the video terminal **950**, the power source terminal **900**, and the imager **906**. The switch **951** connects the power source line **30** and the video signal line **34** to each other in the second period and disconnects the power source line **30** and the video signal line **34** from each other in the first period. The switch **951** connects the imager **906** and the video signal line **34** to each other in the first period and disconnects the imager **906** and the video signal line **34** from each other in the second period.

(119) The switch **951** enters any one of a video output state and a power source output state. The switch **951** can switch between the video output state and the power source output state. When the state of the switch **951** is the video output state, the switch **951** connects the imager **906** and the video signal line **34** to each other and disconnects the power source line **30** and the video signal line **34** from each other. When the state of the switch **951** is the power source output state, the switch **951** disconnects the imager **906** and the video signal line **34** from each other and connects the power source line **30** and the video signal line **34** to each other.

(120) The switching control circuit **905** outputs a switching control signal to the switch **951**, thus controlling the state of the switch **951**. The switching control circuit **905** sets the state of the switch **951** to the video output state in the first period. At this time, the video signal output from the imager **906** is output to the video terminal **904** and is output to the video terminal **950** via the switch **951**. The switching control circuit **905** sets the state of the switch **951** to the power source output state in the second period. At this time, the power source line **30** and the video signal line **34** are short-circuited, and the second power source voltage is output to the video terminal **950**.

(121) The video terminal **950** is connected to the video signal line **34**. The video terminal **950** outputs the video signal or the second power source voltage to the video signal line **34**. The video signal line **34** is a signal line disposed in the transmission cable **3**. For example, the sum of the cross-sectional area of the video signal line **31** and the cross-sectional area of the video signal line **34** is the same as the cross-sectional area of the video signal line **31** shown in FIG. 2. The video signal line **34** transfers the video signal output from the imager **906** to the processor **6e** in the first period. The video signal is transferred to the processor **6e** by both the video signal line **31** and the video signal line **34** in the first period. For example, the video signal transferred to the processor **6e** by both the video signal line **31** and the video signal line **34** may be a differential signal. The video signal line **34** transfers the second power source voltage output from the power source terminal **900** to the processor **6e** in the second period.

(122) The video terminal **651** is connected to the video signal line **34**. The video signal or the second power source voltage transferred by the video signal line **34** is input to the video terminal **651**.

(123) The switch **650** (second switching circuit) constitutes a setting circuit. The switch **650** is connected to the video terminal **651**, the voltage measurement circuit **609**, and the video-signal-processing circuit **607**. The switch **650** connects the video signal line **34** and the video-signal-processing circuit **607** to each other in the first period and disconnects the video signal line **34** and the video-signal-processing circuit **607** from each other in the second period. The switch **650** connects the video signal line **34** and the voltage measurement circuit **609** to each other in the second period and disconnects the video signal line **34** and the voltage measurement circuit **609** from each other in the first period.

(124) The switch **650** enters any one of a video output state and a power source output state. The switch **650** can switch between the video output state and the power source output state. When the state of the switch **650** is the video output state, the switch **650** connects the video signal line **34** and the video-signal-processing circuit **607** to each other and disconnects the video signal line **34** and the voltage measurement circuit **609** from each other. When the state of the switch **650** is the power source output state, the switch **650** disconnects the video signal line **34** and the video-signal-processing circuit **607** from each other and connects the video signal line **34** and the voltage measurement circuit **609** to each other.

(125) The switching control circuit **605** outputs a switching control signal to the switch **650**, thus controlling the state of the switch **650**. The switching control circuit **605** sets the state of the switch **650** to the video output state in the first period. At this time, the video signal transferred by the video signal line **34** and input to the video terminal **651** is output to the video-signal-processing circuit **607** via the switch **650**. The switching control circuit **605** sets the state of the switch **650** to the power source output state in the second period. At this time, the second power source voltage transferred by the video signal line **34** and input to the video terminal **651** is output to the voltage measurement circuit **609** via the switch **650**.

(126) The video signal line **31** (first signal line) and the video signal line **34** (second signal line) are connected in parallel to the camera unit **9e** and the processor **6e**. A setting circuit including the switch **650** and the switch **951** sets the camera unit **9e** and the processor **6e** to be in the first state in the first period and sets the camera unit **9e** and the processor **6e** to be in the second state in the second period. The video signal is transferred to the processor **6e** by the video signal line **31** and

the video signal line **34** in the first state. The load circuit **901** consumes the second power source voltage as a current in the second state. The second power source voltage is output to the video signal line **34** in the second state. The voltage measurement circuit **609** measures a value of the second power source voltage transferred by the video signal line **34**.

(127) In the sixth embodiment, the endoscope system **1e** can monitor a power source voltage provided to the imager **906** as with the endoscope system **1** according to the first embodiment.

(128) The video signal line **31** is not used for transferring the second power source voltage. Therefore, the imager **906** can output the video signal also in the second period.

(129) Since the resistance value of the video signal line **31** is greater than a combined resistance value of the video signal line **31** and the video signal line **34**, there is a possibility that the characteristics of the video signal in the second period are different from those of the video signal in the first period. However, the endoscope system **1e** can transfer the second power source voltage from the camera unit **9e** to the processor **6e** in the second period by allocating a blanking period to the second period or by allocating a period during which the imager **906** outputs a signal not required to be highly accurate to the second period.

Seventh Embodiment

(130) FIG. **9** shows an internal configuration of an endoscope system **1f** according to a seventh embodiment of the present invention. The same configuration as that shown in FIG. **2** will not be described. The endoscope system **1f** shown in FIG. **9** includes a camera unit **9f** and a processor **6f**. The transmission cable **3** shown in FIG. **1** includes a power source line **30**, a video signal line **31**, a control signal line **32**, and a control signal line **35** shown in FIG. **9**. The camera unit **9f** and the processor **6f** are connected to each other by the power source line **30**, the video signal line **31**, the control signal line **32**, and the control signal line **35**.

(131) The camera unit **9f** includes a power source terminal **900**, a load circuit **901**, a video terminal **904**, a control terminal **922**, a switch **930**, and a control terminal **961**. The load circuit **901** includes a switching control circuit **905** and an imager **906**. At least one of the switching control circuit **905** and the switch **930** may be disposed in the imager **906**.

(132) The processor **6f** includes a voltage generation circuit **600**, a voltage-and-current measurement circuit **601**, a power source terminal **602**, a video terminal **603**, a switching control circuit **605**, a video-signal-processing circuit **607**, a resistance adjustment circuit **608**, a voltage measurement circuit **609**, a resistance calculation circuit **610**, a voltage control circuit **611**, a timing generator (TG) **620**, a control terminal **621**, a switch **630**, and a control terminal **661**. All or part of the configuration of the processor **6f** shown in FIG. **9** may be disposed in the operation unit **4** or the connector unit **5** shown in FIG. **1**.

(133) The TG **620** (control signal generation circuit) generates an imager control signal used for controlling the state of the imager **906**. The TG **620** outputs the generated imager control signal to the switch **630** and the control terminal **661**.

(134) The switch **630** (second switching circuit) constitutes a setting circuit. The switch **630** is connected to the control terminal **621**, the voltage measurement circuit **609**, and the TG **620**. The switch **630** connects the control signal line **32** and the TG **620** to each other in the first period and disconnects the control signal line **32** and the TG **620** from each other in the second period. The switch **630** connects the control signal line **32** and the voltage measurement circuit **609** to each other in the second period and disconnects the control signal line **32** and the voltage measurement circuit **609** from each other in the first period.

(135) The switch **630** enters any one of a control signal output state and a power source output state. The switch **630** can switch between the control signal output state and the power source output state. When the state of the switch **630** is the control signal output state, the switch **630** connects the control signal line **32** and the TG **620** to each other and disconnects the control signal line **32** and the voltage measurement circuit **609** from each other. When the state of the switch **630** is the power source output state, the switch **630** disconnects the control signal line **32** and the TG

620 from each other and connects the control signal line **32** and the voltage measurement circuit **609** to each other.

(136) The switching control circuit **605** outputs a switching control signal to the switch **630**, thus controlling the state of the switch **630**. The switching control circuit **605** sets the state of the switch **630** to the control signal output state in the first period. At this time, the imager control signal output from the TG **620** is output to the control signal line **32** via the switch **630** and the control terminal **621** and is output to the control signal line **35** via the control terminal **661**. The switching control circuit **605** sets the state of the switch **630** to the power source output state in the second period. As described later, the control signal line **32** transfers the second power source voltage from the camera unit **9f** to the processor **6f** in the second period. The second power source voltage transferred by the control signal line **32** and input to the control terminal **621** is output to the voltage measurement circuit **609** via the switch **630**.

(137) The imager control signal generated by the TG **620** is input to the control terminal **621** via the switch **630** and is input to the control terminal **661**. The control terminal **621** is connected to the control signal line **32**. The control terminal **621** outputs the imager control signal to the control signal line **32**. The control signal line **32** transfers the imager control signal output from the control terminal **621** to the camera unit **9f**.

(138) The control terminal **661** is connected to the control signal line **35**. The control terminal **661** outputs the imager control signal to the control signal line **35**. The control signal line **35** is a signal line disposed in the transmission cable **3**. For example, the sum of the cross-sectional area of the control signal line **32** and the cross-sectional area of the control signal line **35** is the same as the cross-sectional area of the control signal line **32** shown in FIG. 5. The control signal line **35** transfers the imager control signal output from the control terminal **661** to the camera unit **9f**. The control signal is transferred to the camera unit **9f** by both the control signal line **32** and the control signal line **35** in the first period. For example, the control signal transferred to the camera unit **9f** by both the control signal line **32** and the control signal line **35** may be a differential signal.

(139) The control terminal **922** is connected to the control signal line **32**. The imager control signal transferred by the control signal line **32** is input to the control terminal **922**. The control terminal **922** outputs the imager control signal to the switch **930**.

(140) The switch **930** (first switching circuit) constitutes a setting circuit. The switch **930** is connected to the control terminal **922**, the power source terminal **900**, and the imager **906**. The switch **930** connects the power source line **30** and the control signal line **32** to each other in the second period and disconnects the power source line **30** and the control signal line **32** from each other in the first period. The switch **930** connects the imager **906** and the control signal line **32** to each other in the first period and disconnects the imager **906** and the control signal line **32** from each other in the second period.

(141) The switch **930** enters any one of an imager control state and a power source output state. The switch **930** can switch between the imager control state and the power source output state. When the state of the switch **930** is the imager control state, the switch **930** connects the imager **906** and the control signal line **32** to each other and disconnects the power source line **30** and the control signal line **32** from each other. When the state of the switch **930** is the power source output state, the switch **930** disconnects the imager **906** and the control signal line **32** from each other and connects the power source line **30** and the control signal line **32** to each other.

(142) The switching control circuit **905** outputs a switching control signal to the switch **930**, thus controlling the state of the switch **930**. The switching control circuit **905** sets the state of the switch **930** to the imager control state in the first period. At this time, the imager control signal transferred by the control signal line **32** and input to the control terminal **922** is output to the imager **906** via the switch **930**. The imager **906** generates the video signal in accordance with the imager control signal by using the second power source voltage. The switching control circuit **905** sets the state of the switch **930** to the power source output state in the second period. At this time, the power source

line 30 and the control signal line 32 are short-circuited, and the second power source voltage is output to the control terminal 922.

(143) The control terminal 922 outputs the second power source voltage to the control signal line 32. The control signal line 32 transfers the second power source voltage output from the control terminal 922 to the processor 6f. The second power source voltage transferred by the control signal line 32 is input to the control terminal 621. The control terminal 621 outputs the second power source voltage to the voltage measurement circuit 609 via the switch 630.

(144) The control terminal 961 is connected to the control signal line 35. The imager control signal transferred by the control signal line 35 is input to the control terminal 961. The control terminal 961 outputs the imager control signal to the imager 906.

(145) The control signal line 32 (first signal line) and the control signal line 35 (second signal line) are connected in parallel to the camera unit 9f and the processor 6f. A setting circuit including the switch 630 and the switch 930 sets the camera unit 9f and the processor 6f to be in the first state in the first period and sets the camera unit 9f and the processor 6f to be in the second state in the second period. The imager control signal is transferred to the camera unit 9f by the control signal line 32 and the control signal line 35 in the first state. The load circuit 901 consumes the second power source voltage as a current in the second state. The second power source voltage is output to the control signal line 32 in the second state. The voltage measurement circuit 609 measures a value of the second power source voltage transferred by the control signal line 32.

(146) In the seventh embodiment, the endoscope system 1f can monitor a power source voltage provided to the imager 906 as with the endoscope system 1 according to the first embodiment.

(147) The video signal line 31 is not used for transferring the second power source voltage. Therefore, the imager 906 can output the video signal also in the second period.

(148) The control signal line 35 is not used for transferring the second power source voltage. Therefore, the imager 906 can receive the control signal also in the second period.

(149) Since the resistance value of the control signal line 35 is greater than a combined resistance value of the control signal line 32 and the control signal line 35, there is a possibility that the characteristics of the control signal in the second period are different from those of the control signal in the first period. However, the endoscope system 1f can transfer the second power source voltage from the camera unit 9f to the processor 6f by allocating a period during which the imager 906 receives a signal not required to be fast or highly accurate to the second period.

(150) While preferred embodiments of the invention have been described and shown above, it should be understood that these are examples of the invention and are not to be considered as limiting. Additions, omissions, substitutions, and other modifications can be made without departing from the spirit or scope of the present invention. Accordingly, the invention is not to be considered as being limited by the foregoing description, and is only limited by the scope of the appended claims.

Claims

1. An imaging device, comprising: a camera unit including an image sensor configured to receive a first power source voltage transferred by a power source line as a second power source voltage and generate a video signal, wherein the video signal is output to a video signal line, and a load circuit; a control unit including a signal reception circuit configured to receive the video signal, a voltage generation circuit configured to output the first power source voltage to the power source line, a voltage measurement circuit configured to measure a value of the second power source voltage, and a voltage adjustment circuit configured to adjust a value of the first power source voltage by controlling the power source voltage generation circuit based on the value of the second power source voltage; and a setting circuit configured to set the camera unit to be in a first state in a first period and set the camera unit to be in a second state in a second period different from the first

period, wherein the video signal generated by the image sensor is output to the video signal line in the first state, wherein output of the video signal from the image sensor to the video signal line is stopped in the second state, wherein the load circuit is configured to consume the second power source voltage as a current in the second state, wherein the second power source voltage is output to the video signal line in the second state, wherein output of the second power source voltage to the video signal line is stopped in the first state, and wherein the voltage measurement circuit is configured to measure the value of the second power source voltage transferred by the video signal line.

2. The imaging device according to claim 1, wherein the setting circuit includes: a first switching circuit that is disposed in the camera unit and is configured to connect the power source line and the video signal line to each other in the second period and disconnect the power source line and the video signal line from each other in the first period; and a second switching circuit that is disposed in the control unit and is configured to connect the video signal line and the signal reception circuit to each other in the first period, disconnect the video signal line and the signal reception circuit from each other in the second period, connect the video signal line and the voltage measurement circuit to each other in the second period, and disconnect the video signal line and the voltage measurement circuit from each other in the first period.

3. The imaging device according to claim 2, further comprising a resistance circuit including the voltage measurement circuit, wherein the resistance circuit is configured to have a greater resistance value than a resistance value of the video signal line, and wherein the second switching circuit is configured to connect the video signal line and the resistance circuit to each other in the second period and disconnect the video signal line and the resistance circuit from each other in the first period.

4. The imaging device according to claim 1, wherein the camera unit further includes a transmission buffer configured to enter any one of a third state and a fourth state, wherein, when a state of the transmission buffer is the third state, the transmission buffer is configured to output the video signal generated by the image sensor to the video signal line, wherein, when the state of the transmission buffer is the fourth state, the transmission buffer is configured to stop output of the video signal to the video signal line, and wherein the setting circuit is configured to set the state of the transmission buffer to the third state in the first period and set the state of the transmission buffer to the fourth state in the second period.

5. The imaging device according to claim 1, wherein the voltage adjustment circuit is configured to calculate a resistance value of the power source line based on the value of the first power source voltage, the value of the second power source voltage, and a value of a current that flows through the power source line and adjust the value of the first power source voltage based on the resistance value.

6. An endoscope system, comprising: a scope that has a distal end and is to be inserted into a living body; and the imaging device according to claim 1, wherein the camera unit is disposed in the distal end.

7. An imaging device, comprising: a camera unit including an image sensor configured to receive a first power source voltage transferred by a power source line as a second power source voltage, receive a control signal transferred by a control signal line, and generate a video signal in accordance with the control signal, wherein the video signal is output to a video signal line, and a load circuit; a control unit including a signal reception circuit configured to receive the video signal, a control signal generation circuit configured to generate the control signal and output the generated control signal to the control signal line, a voltage generation circuit configured to output the first power source voltage to the power source line, a voltage measurement circuit configured to measure a value of the second power source voltage, and a voltage adjustment circuit configured to adjust a value of the first power source voltage by controlling the power source voltage generation circuit based on the value of the second power source voltage; and a setting circuit configured to

set the camera unit to be in a first state in a first period and set the camera unit to be in a second state in a second period different from the first period, wherein the control signal is output to the control signal line in the first state, wherein output of the control signal to the control signal line is stopped in the second state, wherein the load circuit is configured to consume the second power source voltage as a current in the second state, wherein the second power source voltage is output to the control signal line in the second state, wherein output of the second power source voltage to the control signal line is stopped in the first state, and wherein the voltage measurement circuit is configured to measure the value of the second power source voltage transferred by the control signal line.

8. An imaging device, comprising: a camera unit including an image sensor configured to receive a first power source voltage transferred by a power source line as a second power source voltage, receive a control signal transferred by a control signal line, and generate a video signal in accordance with the control signal, wherein the video signal is output to a video signal line, and a load circuit; a control unit including a signal reception circuit configured to receive the video signal, a control signal generation circuit configured to generate the control signal and output the generated control signal to the control signal line, a voltage generation circuit configured to output the first power source voltage to the power source line, a voltage measurement circuit configured to measure a value of the second power source voltage, and a voltage adjustment circuit configured to adjust a value of the first power source voltage by controlling the power source voltage generation circuit based on the value of the second power source voltage; and a setting circuit configured to set the camera unit to be in a first state in a first period and set the camera unit to be in a second state in a second period different from the first period, wherein any one of the power source line, the video signal line, and the control signal line includes a first signal line and a second signal line connected in parallel to the camera unit and the control unit, wherein any one of the first power source voltage, the video signal, and the control signal is transferred by the first signal line and the second signal line in the first state, wherein the load circuit is configured to consume the second power source voltage as a current in the second state, wherein the second power source voltage is output to any one of the first signal line and the second signal line in the second state, and wherein the voltage measurement circuit is configured to measure the value of the second power source voltage transferred by any one of the first signal line and the second signal line.
